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1. A database is a place where information is stored and organized so it can be easily found and used.
- * Databases are essential in the modern world because they help businesses and organizations store large amounts of data efficiently and retrieve information quickly when needed.

Examples of where databases are used:

- * Banks - they use databases to store customer accounts and transaction records.
- * Hospitals - they use databases to keep patients records and medical histories.

2. A data warehouse is a large storage system that collects and stores data from different sources in one place for reporting and analysis.

Difference: A database stores current data for daily operations while a data warehouse stores historical data for reporting and analysis.

Why companies use a data warehouse:

- * To analyze data
- identify trends
- make better business decisions

3. Data lake - is a storage system that keeps large amounts of data in its original format.

Difference: A data lake stores raw data, while a data warehouse stores organized and processed data for analysis.

When to use a data lake:

A business uses a data lake when it needs to store large amounts of different types of data for future analysis.

4. A data mart: is a smaller part of a data warehouse that focuses on one department, such as sales or finance. It helps that department access the data they need quickly.

5. Structured data: organised in rows and columns.

e.g Employee database

Semi-structured data: Has some structure but not in tables

e.g JSON file.

Unstructured data: Has no fixed format

e.g Images or videos

6. Data modeling is planning how data will be organised in a database. It helps keep data organised and easy to use.

Conceptual model: Shows the main data and relationships

Logical model: Shows more detail about the data structure

7. Data mining - is finding useful pattern in large amounts of data. Businesses use it to make better decisions.

e.g - stores recommended products based on what customers buy

- Banks detect suspicious transactions

8. A DBMS is software used to create, store, and manage databases

MySQL: Good for websites and small businesses.

PostgreSQL: Good for large business systems

Oracle: used by large companies like banks

MongoDB: Good for apps that use flexible data

9. ETL is a process used to move data into a system

Extract: Get data from different sources

Transform: Clean and change the data

Load: Store the data in a database or data warehouse

10. CSV: Stores data in rows and columns. Used for simple datasets

JSON: Stores data in a structured format. Used in APIs and web ^{apps}

Parquet: Stores large data efficiently. Used in big data systems

Avro: Used for fast data processing and data exchange

11. AI is technology that allows machines to think and act like humans

12. Machine learning is when computers learn from data without being programmed directly.

Types:

- Supervised learning: Learns from labeled data
- Unsupervised learning: Finds patterns in unlabeled data
- Reinforcement learning: Learn by trial and error

13. Deep learning is a part of machine learning that uses neural networks to learn from large amounts of data

14. Natural Language Processing (NLP) is a type of AI that helps computers understand human language.

15. Large Language Model (LLM) is an AI model trained on a large amount of text to understand and generate language

16. Generative AI - is AI that creates new content like text, images or music.
17. Training dataset - is the data used to teach an AI model
18. Computer vision - is AI that helps computers "see" and understand images or videos
19. An API is a tool that allows different software systems to talk to each other.
20. An API key is a special code used to identify and allow access to an API
21. A REST API is a way for systems to communicate
(Representational State Transfer)
22. JSON is a simple format used to store and exchange data
23. Prompt engineering is how you write good instruction for AI tools
24. ~~A~~ map function applies a rule to every item in a list or dataset
25. Cloud computing means storing and using data or programs over the internet instead of your computer
26. A version control system tracks changes in files over time
27. A jupyter Notebook is a tool used to write and run code step by step.
28. Python is a programming language used for data analysis, AI and web development
29. Data privacy is about protecting personal information and making sure it is used properly
30. Data ethics is about using data in a fair and responsible way
31. Data governance is how a company manages and controls its data
32. AI bias happens when an AI system gives unfair results

because of bad or unbiased data

33 Differences between Data roles

Role	What they do	Skills
Data analyst	Analyses data and makes reports	Excel, SQL
Data Scientist	Builds models and predictions	Python, ML
Data Engineer	Builds data systems	Databases, ETL
AI Engineer	Builds AI systems	Deep learning, AI tools

34. Business Intelligence (BI) is the use of data to help businesses make better decisions

35. The most interesting concept was Artificial Intelligence and Machine Learning. ~~It was surprising how new~~

36. The most difficult concept was data modelling and ETL

37. Data and AI will make work faster and smarter. For example, they can help businesses predict customer needs, improve healthcare and automate tasks like reporting and analysis. I think these skills will be important in many careers.

References

1. IBM⁽²⁰²⁴⁾. What is Data Science? <https://www.ibm.com>
2. Oracle⁽²⁰²⁴⁾. What is Database? <https://www.oracle.com>
3. AWS (2024) what is cloud computing? <https://www.aws.amazon.com>
4. W3Schools (2024) - SQL Tutorial. <https://www.w3schools.com>
5. Microsoft (2024). What is AI? <https://www.microsoft.com>
6. GeeksforGeeks (2024). Data Structures & DBMS. <https://www.geeksforgeeks.org>
7. Google Cloud (2024). Machine Learning Basics. <https://cloud.google.com>
8. IBM (2024). What is Data Governance? <https://www.ibm.com>